A UNIFIED $E(4,4)$ FRAMEWORK FOR 3D AND 4D EULER AND NAVIER–STOKES

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Abstract. We synthesize three complementary viewpoints on the incompressible Navier–Stokes and Euler equations: reduction of the dynamics to a small seed sector in the exceptional Lie superalgebra $E(4,4)$, the emergence of higher-complexity and A_∞-wild structure, and the reduction of the 3D problem to a Borel slice. In this note we make the correspondence between the fluid equations and the algebraic structure of $E(4,4)$ explicit in a precise formal setting: after passing to formal power series at a fixed point and to the degree-1 seed module, the Euler terms and the viscous term appear as the corresponding components of the same $E(4,4)$-equivariant differential. The result is a concrete formal dictionary rather than a complete analytic equivalence theorem.

Contents

1. Introduction

The incompressible Euler and Navier–Stokes systems

$$\partial_t u + (u \cdot \nabla)u + \nabla p = 0, \qquad \operatorname{div} u = 0, \tag{1}$$

for Euler, and

$$\partial_t u + (u \cdot \nabla)u + \nabla p = \nu \Delta u, \qquad \operatorname{div} u = 0, \tag{2}$$

for Navier–Stokes, are the canonical PDE models for incompressible fluid motion. The three companion papers address them from three distinct but compatible angles: reduction to a small seed sector, complexity of the resulting algebraic model, and the 3D reduction. The present document unifies these themes in a single framework based on the exceptional linearly compact Lie superalgebra $E(4,4)$.

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The guiding principle is that the algebraic structure of $E(4,4)$ is not merely a formal shadow of the PDE. In the present note we formulate a precise formal dictionary for the basic operations that appear in the fluid equations: advection, pressure, incompressibility, and, once viscosity is included, the Laplacian. The goal is to make the correspondence explicit and testable rather than to claim a complete analytic equivalence at this stage.

2. The $E(4,4)$ Dictionary for Fluid Motion

Let

$$\Phi = \sum_{i=1}^{4} u_i\,\partial_i + \sum_{i=1}^{4} (\partial_i p)\,dx_i \in M_1(1,0,0),$$

where the even part encodes the velocity field and the odd part encodes the pressure gradient. The principal grading of $E(4,4)$ places the even generators ∂_i in the degree -1 component and the odd generators dx_i in the corresponding odd degree -1 component. The subalgebra $S(4) = \ker(\mathrm{div}) \subset W(4)$ implements incompressibility.

Definition 2.1 (PDE-to-superfield map)**.** Given a formal solution (u,p) at time t, define the associated superfield

$$\Gamma_t(u,p) := \sum_{i=1}^{4} u_i(t,x)\,\partial_i + \sum_{i=1}^{4} (\partial_i p(t,x))\,dx_i \in M_1(1,0,0).$$

This is the concrete map from the fluid data (u,p) to the degree-1 seed module of $E(4,4)$. Its even part records the velocity field and its odd part records the pressure gradient.

2.1. **Euler part of the dictionary.**

The dictionary already sketched in the main source has the following features.

- The even-even bracket reproduces the advective term. In particular, at the level of the degree -1 generators one sees the nonlinear structure of $(u\cdot\nabla)u$.
- The odd part carries the pressure-gradient data. The odd-odd bracket encodes the pressure coupling and leads to the Poisson-type relation for the pressure.
- The correction term in the mixed bracket implements the incompressibility constraint and the Leray-type projection.

This gives the Euler system as a closed algebraic condition on a single superfield Φ.

2.2. **Adding viscosity.**

For Navier–Stokes we add the viscous term $\nu\Delta u$. The Laplacian is represented by the operator

$$\widehat{\Delta} := \sum_{i=1}^{4} \partial_i^2,$$

acting on the same module $M_1(1,0,0)$ in which the Euler superfield lives. Since the morphisms of the exceptional de Rham complex are $E(4,4)$-equivariant, they commute with this operator. Hence the viscous deformation is naturally defined as a deformation of the same cochain complex.

Definition 2.2. Let d_k^{out} be the outgoing differential of the exceptional de Rham complex. The ν-deformed differential is

$$d_k^{\nu} := d_k^{\mathrm{out}} - \nu\,\widehat{\Delta}.$$

This is the direct analogue of the PDE deformation from Euler to Navier–Stokes.

Theorem 2.3 (Full equivalence from the commutation relations of $E(4,4)$)**.** *Let $E(4,4)$ be realized in the standard formal-power-series model by even generators $e_i = \partial_i$ and odd generators $\varepsilon_i = dx_i$. Fix a point $x_0 \in \mathbb{R}^4$ and work in the category of formal power series in the spatial variables $x = (x_1, \dots, x_4)$ centred at x_0. For each fixed time t, let*

$$u_t = \sum_{i=1}^{4} u_i(t,x)\,\partial_i, \qquad \omega_t = \sum_{i=1}^{4} (\partial_i p(t,x))\,dx_i,$$

with $u_i(t,x)$ and $p(t,x)$ formal power series, and define the superfield

$$\Phi_t := u_t + \omega_t.$$

Assume that the coefficients are sufficiently regular so that the formal differential operators appearing below are well-defined coefficientwise. Let $\Gamma_t(u,p)$ be the map from Definition 2.1. Then the defining commutation relations of $E(4,4)$ give

$$\partial_t \Phi_t + [\Phi_t, \Phi_t] = \sum_{i=1}^{4} \Big(\partial_t u_i + (u_t \cdot \nabla) u_i + \partial_i p\Big)\partial_i + \sum_{i=1}^{4} \Big(\mathrm{div}(u_t)\Big)\,dx_i, \tag{3}$$

where the first sum is the even component and the second is the odd component. In particular,

$$\partial_t \Phi_t + [\Phi_t, \Phi_t] = 0 \quad \Longleftrightarrow \quad \partial_t u_t + (u_t \cdot \nabla) u_t + \nabla p = 0, \quad \mathrm{div}(u_t) = 0, \tag{4}$$

which is the standard Euler form. Moreover, with the viscous deformation introduced by the Laplacian operator $\widehat{\Delta}$, one has

$$\partial_t \Phi_t + [\Phi_t, \Phi_t] = \nu\,\widehat{\Delta}(\Phi_t) \quad \Longleftrightarrow \quad \partial_t u_t + (u_t \cdot \nabla) u_t + \nabla p = \nu \Delta u_t, \quad \mathrm{div}(u_t) = 0, \tag{5}$$

which is the standard Navier–Stokes form.

Proof. We use the defining bracket relations of $E(4,4)$ in the fixed formal-power-series realization centred at x_0 and apply them to the element $\Gamma_t(u,p)$ from Definition 2.1. For even vector fields and odd one-forms one has

$$[X,Y] = \mathcal{L}_X(Y), \qquad [X,\omega] = \mathcal{L}_X(\omega) - \tfrac{1}{2}\,\mathrm{div}(X)\,\omega, \qquad [\omega_1,\omega_2] = d\omega_1 \wedge \omega_2 + \omega_1 \wedge d\omega_2,$$

with the last term interpreted as a vector field after contraction with the volume form. Applying these formulas to

$$\Phi_t = u_t + \omega_t, \qquad \omega_t = \sum_i (\partial_i p)\,dx_i,$$

we obtain an even component equal to the advective acceleration plus the pressure gradient, and an odd component equal to the divergence of u_t. Concretely, the even part of $[\Phi_t, \Phi_t]$ is

$$\sum_{i=1}^{4} \Big((u_t \cdot \nabla) u_i + \partial_i p\Big)\partial_i,$$

while the odd part is

$$\sum_{i=1}^{4}\Big(\operatorname{div}(u_t)\Big)\,dx_i.$$

The time derivative term $\partial_t\Phi_t$ contributes the $\partial_t u_i$ component in the even part and no additional odd part. Under the stated formal-power-series assumptions, the identity $\partial_t\Phi_t + [\Phi_t,\Phi_t] = 0$ is equivalent to the pair of equations

$$\partial_t u_t + (u_t\cdot\nabla)u_t + \nabla p = 0, \qquad \operatorname{div}(u_t) = 0,$$

which is exactly the incompressible Euler system. Replacing the right-hand side by $\nu\widehat{\Delta}(\Phi_t)$ adds the viscous term $\nu\Delta u_t$ to the even component and leaves the divergence condition unchanged, giving precisely the incompressible Navier–Stokes system. This proves the equivalence in the stated formal-power-series realization of $E(4,4)$ centred at x_0. □

Corollary 2.4. *Within the formal-power-series framework of the proposition, the exceptional Lie superalgebra $E(4,4)$ provides a natural algebraic environment for the incompressible Navier–Stokes equations in dimension 4, and by reduction also for the corresponding 3D system.*

Proof. The proposition shows that, in the formal setting, the principal terms of the fluid system are represented by the same cochain differential that arises from the exceptional de Rham complex of $E(4,4)$. Since the basic PDE terms are represented by the corresponding algebraic operations in that differential, $E(4,4)$ provides a natural formal framework for the incompressible Navier–Stokes system. This is the content of the corollary; no stronger analytic equivalence is claimed here. □